YAMINI SHANKARI AJ

Chennai, India | 9345335161 | Mail
LinkedIn Profile | GitHub Profile | Codolio Profile

SKILLS

Programming Languages	C, C++, Python, Java, JavaScript, TypeScript
Web Development	HTML, CSS, Tailwind CSS, React
Backend and APIs	Node.js, FastAPI, Flask, RESTful APIs
AI/ML	Scikit-learn, TensorFlow, Keras, XGBoost, OpenCV, Pandas, NumPy, YOLO
Databases	MySQL, PostgreSQL, Supabase
Specializations	Computer Vision, Time Series Analysis
Version Control	Git, GitHub, Jira

PROJECTS

Smart Traffic Surveillance System Link

- Constructed an AI-powered ANPR system using **YOLOv8** and **EasyOCR**, achieving **92% plate detection** and **87% OCR accuracy**.
- Engineered a Flask–Supabase PostgreSQL verification pipeline that cross-references recognized plates against **9 vehicle attributes** and performs stolen, suspicious-plate, and violation checks.
- Automated the enforcement workflow from plate recognition to database verification and PDF challan generation, processing each image in an average of **240 ms** through the end-to-end pipeline.

Rubik's Cube Solver Link

- Implemented an ML-powered Rubik's Cube solver using an **SVM with RBF kernel** to classify cube colors from real-time webcam input across all 6 faces.
- Employed OpenCV-based image preprocessing and ROI-based color detection, achieving **96.09% test accuracy** for cube-color classification across **6 color classes**.
- Integrated Kociemba's two-phase solving algorithm with a **Flask REST API** and **React+Vite** interface to generate and visualize step-by-step cube solutions.

Sign Language Interpreter Link

- Created a real-time sign language recognition system using a **3-layer CNN** (TensorFlow, Keras), achieving **85% accuracy** across four gestures.
- Built and annotated a custom dataset of **280 images** and applied OpenCV preprocessing to enhance model performance.
- Trained and validated the 3-layer CNN using an **80/20 data split**, enabling real-time classification of **4 sign language gestures** with live webcam inference.

LEADERSHIP EXPERIENCE

Technical Member — GeeksforGeeks Student Chapter, Chennai, India 2024 – Present

- Mentored peers and junior students in **Data Structures and Algorithms**, competitive programming, and problem-solving by providing guidance on algorithm selection, complexity analysis, coding approaches, and debugging; co-organized **2 coding contests**, contributing to problem-setting, test-case validation, contest coordination and participant support.

Content Creator Lead — Foreign Language Club, Chennai, India 2023 – Present

– Developed German-language educational and promotional content for student outreach, including event announcements and informational materials; coordinated **5+ cultural events** and seminars by supporting event planning, content promotion, social media outreach, and participant engagement, contributing to a **30% increase** in active club membership.

EDUCATION

Bachelor of Technology in Computer Science 08/2023 – 05/2027

SRM Institute of Science and Technology, Ramapuram, Chennai, India **CGPA: 9.17 / 10**

CERTIFICATIONS

Introduction to Database Management System — NPTEL

Introduction to Technology Apprenticeship Job Simulation — Accenture Virtual Experience Program (Forage)

Technology Job Simulation — Deloitte Forage Virtual Experience Program

ACHIEVEMENTS

– **Leadership & Academic Excellence:** Served as Games Captain of the Violet House and ranked **#1 in Computer Science** at the school level.
– **Top 50 Ranking:** Selected among the top 50 teams out of **500+ entries** for **Smart India Hackathon (SIH) 2024 and 2025**.
– **Competitive Programming:** Solved **500+ problems** on popular platforms such as LeetCode, Codechef and GeeksforGeeks.

LANGUAGES

English (Professional)
German (Novice)
Hindi (Beginner)
Tamil (Native)